Deno Jonas Feuerriegel

Mechanical Engineer · Systems Engineering · Sensor Integration · Mechatronics

Hamburg, Germany | deno.feuerriegel@gmail.com | linkedin.com/in/deno-feuerriegel | denofeuerriegel.github.io

Mechanical engineer (B.Sc., grade 1.5) and M.Sc. student with a background as a certified automotive mechatronics technician. Hands-on experience in **independently developing mechatronic systems**, **integrating sensors and electronics** and **validating powertrain software** on HIL test benches and in vehicles. Looking for an entry-level role in systems engineering, sensor integration or mechatronics from **January 2027**.

PROFESSIONAL EXPERIENCE

05/2025 – present

Working Student – Product Development / Unmanned Systems

Projassment e.K.

- Independent concept and development of an **unmanned surface vehicle (USV)** from initial idea to functional prototype
- Mechanical design of parts and assemblies; development and integration of electronic components and custom PCBs
- **Integration of sensors**, propulsion and communication systems (UART); configuration of the open-source autopilot ArduPilot
- Build-up, bring-up and testing of prototypes; iterative optimisation of the overall system

03/2024 – 02/2025

Internship and Bachelor's Thesis – Powertrain Software Development

Volkswagen AG

- Testing, calibration and validation of powertrain software: planned and independently carried out measurement drives with development and prototype vehicles; analysed ECU and sensor signals with ETAS INCA/MDA
- Evaluated measurement data, compared parameter sets and documented and analysed deviations and abnormal software behaviour
- Created, adapted and executed test cases and test plans on the **Hardware-in-the-Loop test bench**
- Bachelor's thesis: built an **end-to-end test automation chain** (Jira, Codebeamer, EXAM, HIL) and integrated it into the existing validation process
- Process and user documentation, onboarding of colleagues; initiated an internal exchange format for interns

11/2021 – 02/2024

Working Student – Test Driver, Vehicle Testing

CoTec GmbH

- Test drives for validation and assessment of **advanced driver assistance systems (ADAS)**
- Vehicle and function assessment; identification and reproduction of functional anomalies and faults
- Documentation and alignment of test results with engineering teams

EDUCATION

04/2025 – present

M.Sc. Sustainable Energy Systems in Mechanical Engineering

HAW Hamburg (University of Applied Sciences)

Current grade average: 1.55 · expected graduation: 12/2026

- Master's thesis (ongoing): technical deployment system for seagrass restoration in the Baltic Sea
- Methods: systems engineering, mechatronic system development, statistical data analysis, machine learning
- Energy: efficient electric drive systems; design, operation and engineering of wind turbines

03/2021 – 03/2025

B.Sc. Mechanical Engineering and Production – Digital Engineering & Mobility

HAW Hamburg (University of Applied Sciences)

Final grade: 1.5

- Bachelor's thesis at Volkswagen AG: automated software commissioning and validation on a HIL test bench
- Technical focus: mechatronics, automation, robotics, autonomous mobile systems, electric drives
- Digital focus: machine learning, data analysis, applied AI, finite element method

08/2017 – 01/2021

Apprenticeship – Automotive Mechatronics Technician

Kuhn + Witte GmbH & Co. KG

- Maintenance, diagnosis and repair of mechanical, electrical and electronic vehicle systems (VW, Audi, Škoda)
- Systematic fault diagnosis with diagnostic tools, measuring equipment, wiring diagrams and OEM information; bus systems (CAN, LIN, MOST, FlexRay)
- Wheel alignment as well as measurement and **calibration of ADAS sensors**

2016

Abitur (university entrance qualification)

Gymnasium Winsen (Luhe)

SELECTED ACADEMIC PROJECTS

B.Sc.
Student project
Grade: 0.7

Digital PID Control Laboratory Experiment

Measurement and Control Engineering

- Designed a hands-on lab experiment introducing PID control; **permanently adopted into the lab curriculum**
- Digital controller on an Arduino Due; electronics developed with LTspice and KiCad

B.Sc.
Project

ROS-Based SLAM and Navigation

Autonomous Robotics

- Integrated and configured a SLAM system (ROS, hector_slam, LiDAR) for a Mecanum-wheel robot incl. navigation, path planning and obstacle avoidance

M.Sc.
Team project
Grade: 0.7

Wind Farm Power Curve and Power Forecasting

CC4E wind farm Curslack

- Analysis and cleansing of operational and met-mast data; filters for a normalised power curve
- Power forecasts from live met-mast and weather data: **$R^2 > 0.85$**, MAE approx. 500 kW at 12.6 MW rated capacity

SKILLS

Embedded & Sensors	Arduino; ArduPilot; ROS; UART; KiCad; LTspice; sensor integration
Test & Validation	Hardware-in-the-Loop; test automation (EXAM); verification and validation; test planning; root-cause analysis; vehicle testing
Automotive	ETAS INCA/MDA; DiagRA; ODIS; CAN/LIN; ADAS; powertrain software
CAD/CAE & PLM	PTC Creo Parametric; Autodesk Fusion; ADINA/FEM; PTC Windchill
Programming & Data	Python; MATLAB/Simulink; Structured Text / PLC; Git; data analysis; machine learning
Tools & Methods	Jira; Confluence; PTC Codebeamer; Microsoft Office; system integration; product development; project management; technical documentation

LANGUAGES & INTERNATIONAL EXPERIENCE

Languages	German – native; English – business fluent
Abroad	2017: three-month stay in the USA
Grading note	German grading scale: 0.7–1.0 = excellent (best), 4.0 = pass